MIL-HDBK-696
30 March 1965

MILITARY HANDBOOK

PLASTIC COATING COMPOUNDS, STRIPPABLE, HOT AND COLD DIP

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DEPARTMENT OF DEFENSE
WASHINGTON, D. C.

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MIL-HDBK-696
Plastic Coating Compounds, Strippable, Hot and Cold Dip
30 March 1965

1. This standardization handbook was developed by the Department of Defense in accordance with established procedures.

2. This publication was approved on 30 March 1965 for printing and inclusion in the military standardization handbook series.

3. This handbook will provide information to formulators and users of hot and cold dip, strippable, plastic coating compounds for military packaging applications. The handbook is not intended to be referenced in purchase specifications except for informational purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to reflect the latest information on plastic coating compounds, strippable, hot and cold dip. It is the intent to review this handbook periodically to ensure its completeness and currency. Users of this document are encouraged to report any errors discovered and any recommendations for changes or inclusions to U. S. Army Materials Research Agency, Watertown, Massachusetts 02172. Additional technical information on these materials may be obtained from the U. S. Army Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland 21005.

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PLASTIC COATING COMPOUNDS, STRIPPABLE, HOT AND COLD DIP

1. INTRODUCTION

1.1 Plastic coating compounds, strippable, hot and cold dipping, are protective, corrosion resistant packaging media that are covered by MIL-P-149 and MIL-P-45021 respectively. These compounds are fluid liquids at dipping temperatures. Objects are coated with these compounds by being immersed for several seconds and withdrawn. The film that coats the withdrawn object is cooled sufficiently to congeal in a matter of a few seconds, thus allowing the coated object to be handled soon after dipping without damage to the film. Oversized objects may be coated by spray techniques using cold dipping compound (MIL-P-45021).

1.2 The development of plastic coating compound, strippable, hot dipping, was necessary in order to meet the military needs for proper maintenance and preservation of military supplies. Modifications have been made since the original development of the compound (meeting the requirements of AXS-1167) to meet the changing military needs and to utilize the advances in the plastic industry.

1.3 Among several desirable improvements in these hot dipping compounds, it became apparent through field experience that sub-zero flexibility, shock resistance and longer durability under weather exposure were urgently needed. The sub-zero shock resistance was particularly urgent, in view of the fact that a good many critical items were being transported at high altitudes where temperatures of -65°F. are frequently encountered.

1.4 In order to make possible the packaging of temperature sensitive items, compounds that could be applied with moderate heat only were developed. These compounds meeting the requirements of MIL-P-45021 were further improved, to render them resistant to fungal attack.

2. COMPOSITION

2.1 Hot dipping (2, 3). Formulations meeting the requirements of MIL-P-149 generally consist of ethylcellulose or cellulose acetate butyrate, together with mineral oil, plasticizers and small quantities of other modifying agents such as waxes, resins and antioxidants. They are classified as type I or type II depending on whether ethyl cellulose or cellulose acetate butyrate is used. A general example of a type I compound may consist of:

Percent by weight
25% ethyl cellulose
65% mineral oil
5% plasticizer
5% resins, waxes and antioxidants

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A general example of type II compound may consist of:

40% cellulose acetate butyrate
55% plasticizer
5% rust inhibiting oil, waxes and antioxidants

A group of recently developed type II formulations incorporating the best feasible weather and sub-zero properties, are listed in table I.

Table I - Composition of type II formulations

Formula	Percent by weight				
	A	B	C	D	E
Cellulose acetate butyrate (EAB 500-5)	42.8	42.8	42.8	42.8	42.8
Polyethylene glycol di-2-ethylhexoate (Flexol 4G0)	47.6				23.8
Di-2-ethylhexyl adipate		47.6		23.8	18.9
Di-octyl sebacate			47.6	23.8	
Dibutyl sebacate					4.9
Zerice (neutral oil)	8.9	8.9	8.9	8.9	8.9
Antioxidant	0.5	0.5	0.5	0.5	0.5
Paraflow	0.2	0.2	0.2	0.2	0.2

2.2 Cold dipping. Materials formulated to meet the requirements of MIL-P-45021 consist of cellulose acetate butyrate combined with various plasticizers, resins, inhibitors, fungicides and volatile solvents. An acceptable formulation which has met the requirements of the specification is as follows:

	Percent by weight
Cellulose acetate butyrate	18.5
Chlorinated biphenyl	15.7
Polyethylene glycol di-2-ethylhexoate	3.6
Mineral oil (Fractol A)	2.4
Xylol	40.4
Isopropyl alcohol	18.2
Phenylsalicylate	0.3
Trilauryl phosphate	0.3
Di-2-ethylhexylamine	0.3
Solubilized 10% copper-8-quinolinolate	0.3

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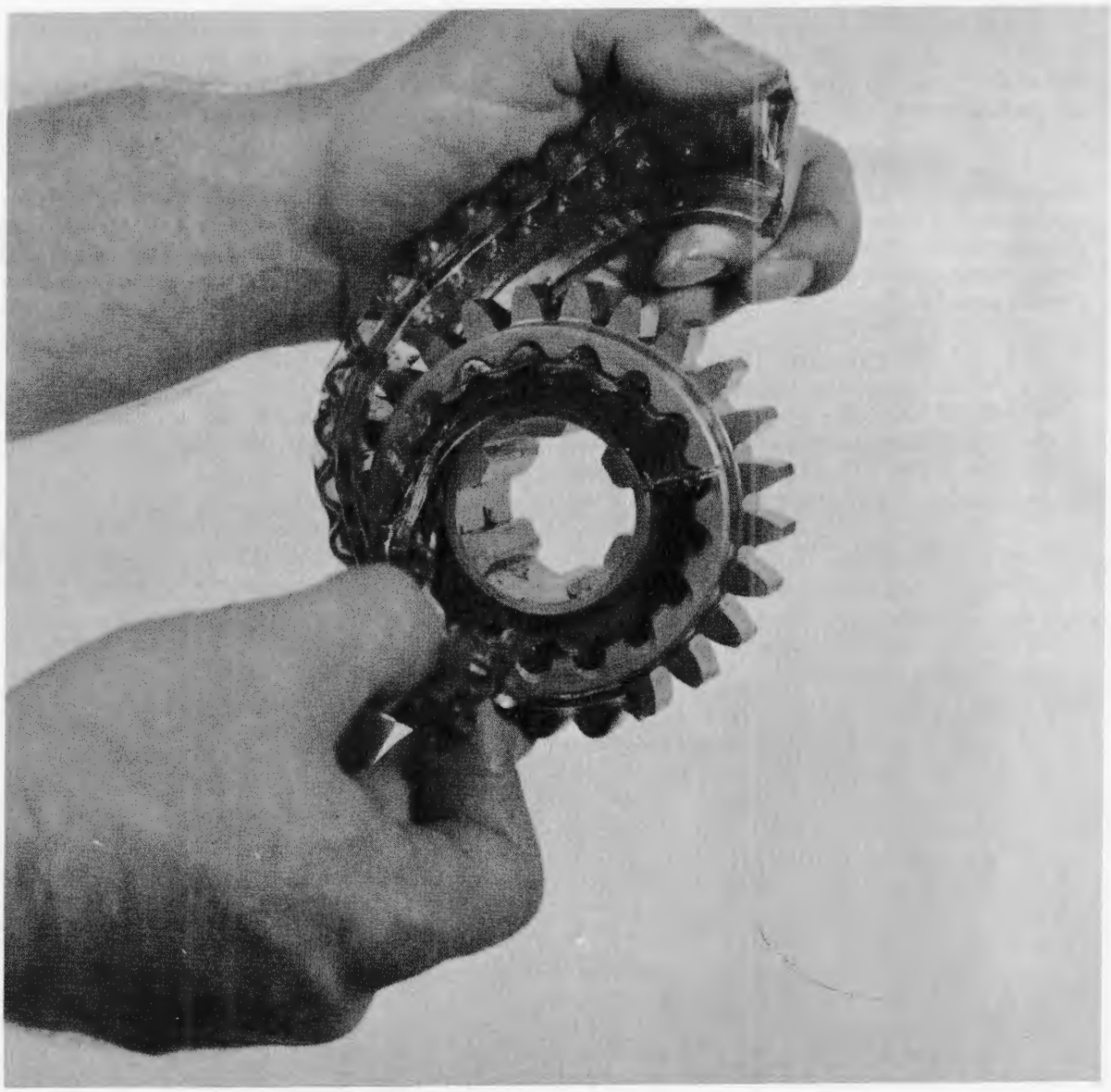


Figure 1. Partially stripped transmission gear coated with Plastic Coating Compound, Strippable, Hot-Dipping.

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3. RAW MATERIALS

3.1 Plasticizers (1, 4, 12, 16). A plasticizer is a low melting solid or high boiling organic liquid. It is added to the polymeric material to extend or modify the natural properties of the resin and to impart new improved properties such as flexibility and weather resistance, to the finished film.

Theoretically the perfect plasticizer is absolutely nonvolatile, but in practice a certain volatility is accepted if other properties justify such acceptance. In addition, it should be compatible, light fast, water insoluble, nonflammable, nontoxic and free from objectionable odor. As with the volatility requirement a choice usually has to be made which will best fill most of these requirements.

3.1.1 Properties of plasticizers. The physical properties of some of the common cellulose ester plasticizers are listed in table II. An important plasticizer for hot dipping compounds is castor oil. Castor oil is mainly a mixed glyceride in which the following acids are combined: ricinoleic 80%, oleic 9%, linoleic 3% and stearic 3%.

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Table II - Properties of plasticizers for cellulose ester(1)*

Plasticizer	Mole. Wt.	Melting Pt.°C	Boiling Pt.°C	Specific Gravity	Refractive Index	Water Solubility %	Free Acidity
Amyl phthalate	306	-55	342	1,022	1,488	0.010	0.27
Benzophenone	182	48	305	1.111	1.602	0.006	0.05
Butyl phthalate	278	-35	325	1.050	1.490	0.013	0.14
Butyl phthalyl butyl glycollate	336	-45	219(5mm.)	1.097	1.488	0.018	0.70
Butyl sebacate	314	---	345	0.933	1.493	0.005	1.05
Butyl succinate	230	-19	255	0.974	1.428	0.023	0.90
Butyl tartrate	262	22	312	1.098	1.445	0.485	2.75
Camphor	152	178	204	0.990	-----	0.170	0.60
O-Cresyl-p-toluene sulphonate	262	52.5	---	1.207	1.558	0.003	0.04
Cyclohexyl-p-toluene sulphonamide	253	86	---	-----	1.542	0.006	0.46
Diethylene glycol dipropionate	218	-82	165(4mm.)	1.056	1.429	3.60	0.52
Ethoxyethyl adipate	290	-70	---	1.036	1.439	0.248	3.64
Ethyl benzoyl benzoate	254	58	355	1.155	1.573	0.010	0.30
Ethyl phthalate	222	-0.3	295	1.118	1.499	0.150	0.07
Ethyl phthalyl ethyl glycollate	280	-35	190(5mm.)	1.177	1.498	0.050	0.50
Ethyl sebacate	258	1.3	308	0.965	1.435	0.018	0.52
Ethyl succinate	174	-22	218	1.048	1.419	0.190	0.09
Ethyl tartrate	206	17	280	1.199	1.445	50	1.01
Ethylene glycol diacetate	146	-78	186	1.128	1.414	1.815	0.14
Ethylene glycol dibutyrate	202	-80	240	1.024	1.424	0.050	9.16
Ethylene glycol dipropionate	174	-80	211-	1.054	1.419	0.160	0.88
Methoxyethyl phthalate	282	-60	209-61 (20mm.)	1.17	1.500	0.838	1.40
Methyl benzoyl benzoate	240	52	351	1.190	1.587	0.005	3.20
Methyl phthalate	194	0	285	1.193	1.513	0.305	0.20
Methyl phthalyl ethyl glycollate	266	-35	189(5mm.)	1.220	1.503	0.053	2.30
Propyl phthalate	250	-31	158(4mm.)	1.071	1.494	0.015	0.65
o- & p-toluene ethyl sulphonamide	199	18	---	1.190	1.540	0.140	4.30

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Table II - Properties of plasticizers for cellulose ester (cont'd)

Plasticizer	Mole Wt.	Melting Pt.°C	Boiling Pt.°C	Specific Gravity	Refractive Index	Water Solubility %	Free Acidity
Triacetin	218	-40	259	1.161	1.429	0.685	1.84
Tributyl phosphate	266	-20	289	0.976	1.421	0.033	3.60
Tributyrin	302	-70	287	1.032	1.434	0.010	0.44
Tricresyl phosphate	368	-35	295(13mm.)	1.175	1.556	0.008	0.55
Triethylene glycol diacetate	235	-60	300	1.111	1.437	50	2.12
Triethylene glycol dibutyrate	290	-70	154-60 (2mm.)	1.033	1.437	0.800	0.60
Triethylene glycol dipropionate	262	-60	138-42 (2mm.)	1.066	1.436	6.70	3.10
Triphenyl phosphate	326	48.5	245(11mm.)	1.185	1.563	0.002	0.21
Tripropionin	260	-50	177-82 (20mm.)	1.078	1.431	0.313	0.42

*Number refers to list of references in bibliography.

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3.1.2 Beyond meeting basic requirements, a number of plasticizers are utilized to add specific properties to the coating. Polyethylene glycol di-2-ethylhexoate is especially useful in both hot dipping and cold dipping compounds to impart low temperature flexibility and strippability. The properties of this plasticizer and others used in table I are listed in table III.

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Table III - Properties of plasticizers used in table I formulations

Characteristic	Polyethylene glycol di-2-ethylhexoate	Di-2-ethylhexyl adipate	Dioctyl sebacate	Dibutyl sebacate
Specific gravity at 20/20°C	0.989	0.927	0.911	0.936
Average weight/gal. at 20°C	8.2	7.73	7.6	7.8
Boiling range at 5 mm Hg., °C	215-290	205-220	256	200
Absolute viscosity at 20°C, cp	25.1	15.3	20-30	7.9
Pour point, °C	-55	-69	-48	-11
Flash point (Cleveland Open Cup) °C	202	192	204	178
Refractive index at 20°C	1.447	1.447	1.451	1.442
Coefficient of expansion at 20°C	0.00081	0.00078		0.00078
Solubility in water at 20°C, % by wt.	Insol.	Insol.	Insol.	0.1
Solubility of water in at 20°C, % by wt.	1.4	0.2		0.2

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3.1.3 The efficiency of the plasticizer is judged on the basis of elongation, flex, temperature, tensile strength, hardness, strippability and weather resistance imparted to the finished film.

3.2 Inhibitors. No single inhibitor, for cold dipping compounds has been found that will offer simultaneous corrosion protection to steel, copper, and
13 aluminum. For the cold dip formulation given in paragraph 2.2, a combination of alkyl phosphate and amine will protect all three metals. Either trioctyl phosphate or trilauryl phosphate may be employed and either di-2-ethylhexylamine or methyl morpholine may be used in conjunction with these organic phosphates. The phosphates may be obtained commercially or prepared directly as described in (23).

3.2.1 For hot dipping compounds, 2,2'-methylenebis (4-methyl-6-tertiary butyl phenol) will protect all three metals in five of the suggested formulations.

3.3 Release agents. Release agents are compounds that are added to the formulation to increase the strippability properties. Cold dipping formulations containing soybean lecithin have been found to have excellent stripping properties. However, there is some detraction from the corrosion protection when it is incorporated into the formulation. Since alkyl phosphates greatly increase the strippability, formulations containing these inhibitors do not require a release agent as such. Silicone oils also serve as release agents.

The large percentage of oils in type I hot dipping compounds, eliminate the need of a release agent in these formulations. The incorporation in type II compounds of oils and combinations of adipates and sebacates impart adequate strippability properties.

3.4 Fungicides. One of the uses for which cold dipping protective compound was developed, was for a fungistatic protective coating for unmounted lens elements (see figure 2). One of the things to be considered in the fungus resistance of cold dipping strippable films applied to lenses and prisms is that if fungi can utilize any constituents of the protective compound, there is a possibility that some of the metabolic products which may be acidic or alkaline, depending on the food source, will diffuse through the film and etch the optical glass.

3.4.1 A good fungicide should have several basic properties. First and most obvious it should completely prevent fungus growth. Secondly, it should prevent changes in mechanical properties due to microbial attack. Third, it should not alter the other protective properties of the film, and finally, it should be approved with regard to toxicological and possible skin irritating properties.

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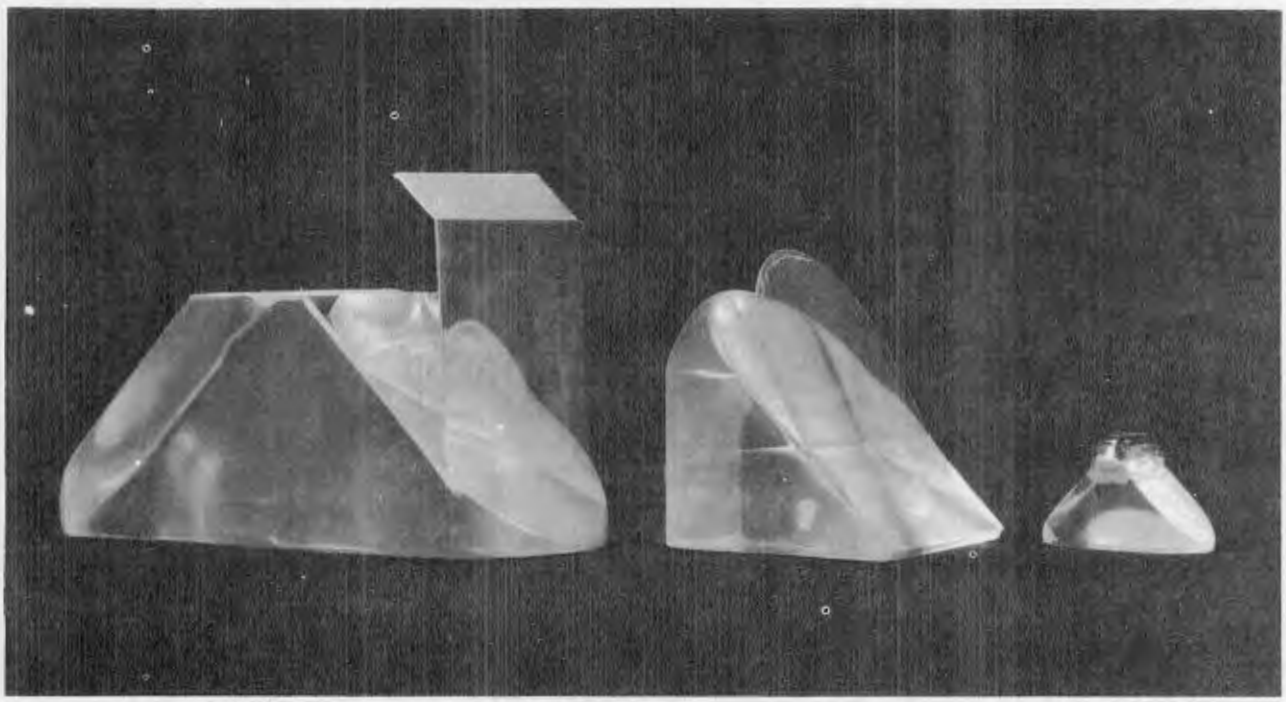


Figure 2. Optical elements coated with Plastic Coating Compound, Strippable, Cold Dipping. Note partial stripping.

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3.4.2 Of the fungicides evaluated to date, the one which most closely meets the above requirements is solubilized copper-8-quinolinolate. The effectiveness of this compound is based on the fact that it dissociates to form copper ions and 8-hydroxy-quinoline, each of which performs differently but uniquely related functions in preventing the growth of fungi. The 8-hydroxy-quinoline inhibits the growth of fungi by precipitating the elements which are essential for the metabolism of the organism. The copper ions function synergistically with the 8-hydroxy-quinoline to provide the degree of fungistatic activity attributed to copper-8-quinolinolate.

3.4.3 Little work has been carried out on the fungistatic evaluation of hot dip compounds or the development of a fungicide for these compounds. A fungistatic barrier may be established by dipping the hot dip coated article into cold dip compound.

3.5 Ultraviolet inhibitors for cellulose acetate butyrate coatings (5). An ultraviolet inhibitor must have a high absorption capacity for the ultraviolet region of the solar spectrum (2900-4000A°). It should cover this entire range without introducing color in the coating. It should dispose of the absorbed energy in such a manner that the cellulose ester chains are not harmed. The inhibitor should have many of the desired characteristics of plasticizers. These include compatibility, permanence, water insolubility, freedom from odor and from color forming on exposure. Finally in getting rid of the absorbed energy, the inhibitor itself should not be physically damaged.

3.5.1 Without inhibitors these coatings on exposure to outdoor conditions bleach, surface-craze, become brittle and finally disintegrate. Previous work has shown that outdoor degradation is primarily the result of photochemical action and is easily followed by intrinsic viscosity measurements. The introduction of 1 percent phenyl salicylate into cellulose acetate butyrate compositions has largely overcome many of these deficiencies.

3.5.2 Other ultraviolet inhibitors. Because phenyl salicylate (salol) is an effective weathering agent for cellulose ester plastics, the isomers of this compound and their methyl ethers have been studied. Ultraviolet absorption characteristics of the compounds have been measured and weathering characteristics of coatings containing these compounds have been determined. Compounds which to date indicate to be most effective are m-hydroxyphenyl benzoate, p-hydroxyphenyl benzoate, phenyl o-hydroxybenzoate and o-hydroxyphenyl benzoate. Those susceptible to quinone formation tend to become yellow on exposure. Those with methoxy substituents are less effective than the corresponding hydroxy compounds. High optical density in the ultraviolet range after irradiation seems to be an essential requirement but does not necessarily assure the material's being an effective inhibitor.

3.6 Anti-blooming agents. The blooming of copper-8-quinolinolate in polymeric compounds has been checked by the use of compatibilizing agent such as the condensation product of formaldehyde and toluene sulfonamide (U. S. Patents 2,689,837 and 2,689,838).

Abiates are also useful in preventing the crystallization of the copper-8-quinolinolate.

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3.7 Solvents. The cold dip solvent usually consists of an aromatic hydrocarbon such as toluene, xylene or a commercial aromatic mixture of comparable boiling range, together with the proper amount of aliphatic alcohol (of three to five carbon atoms). Isopropyl alcohol of 99.7% purity is the most useful alcohol. When the cold dip is applied as a spray, a less volatile fluid is required in order to minimize condensation due to rapid evaporation of the alcohol. Amyl alcohol is satisfactory for this purposes. The solvent-solid ratio lies between 3 to 1 and 7 to 1. The viscosity range should be adjusted in the range of 1500-1600 centipoises. The particular viscosity will depend on the thickness of the coating desired, and the item that is being dipped (see recommended procedure). 16

4. PERFORMANCE CHARACTERISTICS

4.1 Exposure of plastic coating compound, strippable (hot dipping) to salt water, surf and sand action. Hot dip, MIL-P-149, type I and type II, compounds will give adequate corrosion protection against salt water, heavy surf and sand action for more than fifteen days but less than forty-five. At the end of fifteen days exposure, type I compound cannot be stripped properly. Type II compound strips without difficulty at the end of a hundred days exposure. Both type I and type II compounds will give adequate corrosion protection for over a hundred days in sheltered clean sandy beach areas free from heavy seas. Items coated with type I and type II compound are severely corroded and discolored when exposed to organic matter in bottoms having soft silt.

It is recommended that coated parts which become immersed in sea water be removed as expeditiously as possible, as serious loss of corrosion protection can occur within fifteen days.

4.2 Weather exposure of plastic coating compound, strippable (hot dipping). Items coated with type I compounds fail to strip after sixty days of exposure regardless of climatic conditions. Copper and steel items coated with type I compound show corrosion after sixty days. Aluminum items coated with either type I or type II compounds show no signs of corrosion regardless of climatic conditions and after prolonged periods of exposure. Type II compounds strip regardless of exposure time or climate and give adequate protection to copper and steel items in the tropics for 100 days and over 500 days in temperate climates.

It is recommended that type II, Plastic Coating Compound, Strippable, (Hot Dipping) meeting MIL-P-149, be used in preference to type I when coating parts that will be stored for prolonged periods outdoors in temperate and arctic climates and for periods not to exceed three months in damp tropical climates.

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4.3 Platform storage exposure of plastic coating compound, strippable, (hot dipping). Steel and copper items coated with type I compounds develop corrosion after 12 months exposure, and fail to strip after 18 months. Aluminum items coated with either type I or type II are protected indefinitely. The best type II gives steel and copper protection for as long as 1400 days. Type II is recommended for use in preference to type I when coating parts that will be exposed to outdoor platform storage (see figure 3).

The accelerated laboratory test incorporating an oxygen bomb in conjunction with humidity cabinet, as described in MIL-P-149, gave the best correlation with platform storage.

4.4 Weather exposure of plastic coating compound, strippable, (cold dipping), MIL-P-45021. Steel, copper and aluminum items coated with cold dipping compound will remain unchanged after 72 hours exposure in an oxygen bomb, and subsequent exposure for 1440 hours in the 100% relative humidity cabinet at 100°F. To date they have been found to remain unchanged for a minimum period of one year outdoor platform storage and for 18 months indoor storage. If the correlation of outdoor exposure and combination oxygen bomb-humidity test for hot dip compounds is found applicable to cold dip formulations, the outdoor platform storage protection of this type of film may be as high as two to three years. Data on exposure of this type compound to various climatic conditions has not yet been obtained.

4.5 Protection against fungal attack. All cold dip formulations containing copper-8-quinolinolate in concentrations of at least 0.5% based on plasticizer content, are completely fungus resistant (evaluated according to procedure described in MIL-P-45021). Increasing the amount of plasticizer produces a yellow oily film during the test. The protection of other formulations is shown in (23).

4.6 Ultraviolet stability. The cold dip compound at a thickness of 0.048 inch is still soft and pliable after 1000 hours exposure in a weatherometer. Experiments indicate that in addition to the concentration of the ultraviolet inhibitor, the film thickness also has an effect on ultraviolet resistance (5).

4.7 Moisture - vapor permeability. Formulations which do not contain resins and waxes yield coatings which have good water resistance but an average moisture-vapor permeability of about twenty times that of moisture proof cellophane. Formulations containing cellulose ester, plasticizer, resin and wax, produce coatings with permeability approaching that of moisture proof cellophane.

Moisture vapor permeability of hot dip cellulose acetate butyrate based compositions may be varied considerably. In general, moisture resistance is favored by high butyrate content. Typical average permeability values for type I and II are given in table IV.

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Figure 3. Platform storage exposure of panels coated with Plastic Coating Compound, Strippable.

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Table IV - Moisture-vapor permeability (100°F and 95% R. H.) in grams water/100 square inches

Time (hours)	Type I	Type II
24	0.700	0.175
48	0.730	0.195
66	0.830	--
94	--	0.215

5. METHODS OF APPLICATION

5.1 Procedure for use of Hot Dip Strippable Compound.

5.1.1 Description of method. The method consists of coating the prepared items with material conforming to MIL-P-149, type I or type II. Type II compound is required for preservation method IB of MIL-P-116. After the compound sets, the item is wrapped, boxed or canned to retain oil which exudes during storage. The same purpose may be served by giving the wrapped item an overcoat of cold-dip material conforming to MIL-P-45021.

5.1.2 Static load. Coated items must be blocked, braced or cushioned to insure that static load on the compound does not exceed 30 pounds per square inch. When required by contract or order it shall pass "rough handling test" and "cyclic exposure test" (see 7.2).

5.1.3 Dipping equipment. Tanks are described in MIL-T-13291, Tanks, Hot Dip, Directly and Indirectly Heated.

5.1.4 Cleaning of items. Prior to dipping, the article should be thoroughly cleaned in accordance with MIL-P-116.

5.1.5 Suspension of items.

5.1.5.1 String. String has been used as a suspension medium for dipping spare parts in plastic coating compound in the field. It has been found that on parts packaged in this manner, corrosion sometimes occurs where the string and metal are in contact. It is very important therefore that the string or thread be rendered water repellent. The requirements for a high quality non-wicking, water repellent string for use in connection with dipping metallic objects are described in MIL-S-12091.

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5.1.5.2 String dehydration. Dehydration of commonly used string material (cotton, flax, linen and spun glass) may be accomplished in the following manner:

Place the string material in an oven heated to a temperature of 160°F for a period sufficient to completely remove all traces of moisture (usually 2 to 6 hours, depending on the volume of material). One satisfactory method is to cut and loop string material into desired lengths and then dehydrate. Immediately after dehydration, the cord material is impregnated by immersion in heated preservative compound conforming to the requirements of MIL-C-11796. A check to determine if the cord is sufficiently dry is made by dipping a representative sample of the dehydrated cord into strippable compound. Absence of bubbles underneath the compound is indication that the cord is dry.

5.1.5.3 String requirements. The object shall not weigh more than 1/4 of the breaking strength of the suspension cord if it is to be used as a single strand.

5.1.5.4 Metallic wire. Any metallic wire soft enough to permit the formation of a tie (such as copper, aluminum or steel) is also satisfactory as a suspension medium. Metallic foil or ribbon is not satisfactory because it does not permit a sufficient thickness of the strippable compound to be deposited and subsequent handling causes the film to peel from the foil.

5.1.6 Material control.

5.1.6.1 Dipping temperature. Care should be taken so as not to overheat the compound during either melt down period or actual dipping time. Prolonged overheating causes the breakdown or degradation of either cellulose acetate butyrate or ethyl cellulose due to oxidation. Dipping temperatures: type I compound 350-375°F, type II compound 325-350°F.

5.1.6.2 Thickness. Sufficient thickness for a satisfactory coating of strippable protective compound is between 0.050 and 0.100 inch. Thinner coatings are acceptable on small light parts; but in no case shall the minimum thickness be less than 0.025 inch on any surface. Coating thickness is determined by the dipping temperature, the immersion time, the mass or heat capacity of the part, characteristics of the compound and withdrawal time. While a part is being dipped, it takes some of the heat from the melted plastic, and the longer the duration of the dip, the closer the temperature of the part approaches the temperature of the plastic and the thinner the protective compound obtained. Conversely, a massive part when dipped for the same interval will retain a temperature considerably lower than the temperature of the plastic, resulting in greater cooling action on the plastic during the "drain off" period. This results in a thicker protective coat.

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5.1.6.3 Make-up material. When cold compound is added to compensate for that removed by dipping, the addition must be made frequently and in relatively small quantities to prevent wide temperature fluctuation in the tank. If non-interrupted production is required, plastic may be pumped in from a pre-heated tank which melts the plastic and brings the temperature close to that of the dipping tank. When not in use, tanks may be kept at idling temperatures from 250° to 275°F during the work day. A range of 150° to 250°F for overnight holding is recommended.

5.1.6.4 Reuse. The nature of the plastic permits it to be reused a number of times and dipped tailings, stripped coatings, drip etc. can be reused, provided they are not contaminated with oil, grease or dirt. Under no circumstances should plastic of any other formula be combined with the material. Neither shall compounds furnished by different manufacturers be mixed. Impure or unlike compounds may cause phase separation or loss of physical properties required of the material.

5.1.7 Plugging of cavities. Parts with cavities having diameters between 1/4" and 1-1/4" should be plugged with either Corrosion Preventive Compound, Petrolatum, Hot Application, MIL-C-11796, class 1 or class 2, before dipping. By this means many parts can be preserved with dipping compound which would ordinarily require other treatments.

5.1.8 Dipping procedure.

5.1.8.1 Single dip. Part is suspended from string and completely coated in one dip. Dipping time should be as short as possible and ordinarily should not exceed 10 seconds.

The draining positions should prevent pocketing and should yield even and continuous coating. Drainings of any excess compound must not be allowed to flow back into the dipping section of the tank. No dipping should be done in a melted compound which shows air bubbles. If a compound leaves air bubbles in the coating after the part has been dipped, it is unsatisfactory for use.

5.1.8.2 Two step overlap dip. This method is used to coat large parts, or parts shaped so that a satisfactory coating cannot readily be obtained by a single dip. Clean cloth or rubber gloves are worn to protect the parts from finger prints and the parts are dipped by hand. First, one end of the item is dipped, and the item is supported by its undipped end until the initial coating sets. The part is reversed and dipped so that a suitable overlap of at least one inch is formed.

5.1.8.3 Aluminum foil wrap and strippable compound. This method consists of the application of strippable compound over a conforming wrap of aluminum foil (MIL-A-148), applied so as to exclude all possible air pockets. The function of the conforming wrap may serve three purposes as follows:

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(a) To prevent the contact grease or oil preservative from being dissolved by the hot dip compound.

(b) To protect the strippable compound from sharp edges.

(c) To prevent the oil exuded by the compound from contacting the packaged parts. 22

The strippable compound may be applied either by the single dip or two step overlap method. If the aluminum foil is used to protect the plastic, it is recommended that a double thickness of aluminum foil be placed over the sharp edges of the item.

5.1.9 "Setting-up" of the compound. Parts are suspended by their strings or wires until the coating has set and cooled sufficiently so that handling will not cause pressure thinning or cutting of the protective coating. As is the case with all thermoplastics, "setting-up" or hardening is accelerated by forced air or cold air. Water or other liquids should not be used to accelerate setting. After setting, the string or wire should be clipped flush with the outer surface of the coating. In order to insure that no moisture will enter the exposed ends of the wire or string, a low temperature iron is applied and a small amount of plastic is run over the exposed ends.

5.1.10 Depreservation. Slit the coating with the thumb nail or non-metallic instrument such as a wooden chisel and peel like a banana. Solvents or gasoline are not necessary. The oily film remaining on the part contains a rust inhibitor and does not have to be wiped off. Care should be exercised when slitting the film on a critical surface to prevent scratching the surface (see figures 1 and 2).

5.1.11 Solid contaminants and cleaning of tanks. A screen in the hot-dip strippable tank is desirable to screen out carbonized particles and other solid contaminants, if and when they occur. The use of a solvent carbon loosener (P-C-111), or trichloroethylene is recommended for cleaning the tanks in the event the compound overheats.

5.1.12 Inspection standards.

5.1.12.1 At least once during each operating shift, samples of representative parts should be taken for strippability tests and for degradation observations. Coated parts should be allowed to stand at 18 to 26°C (65-80°F) for 6 hours. The coating should then be compared for color with an original standard taken during the first 1/2 hour of operation of each manufacturers lot (or batch) of strippable compound. In addition the part set aside should be stripped, and must show no adhesion to any surface of the parts.

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5.1.12.2 An occasional part should be kept at room temperature for 168 hours and then stripped. If adhesion occurs, checks should be made to determine the cause. If there is any indication that the compound has degraded, as evidenced by a change in color or adhesion upon stripping, dip a flat panel in the compound for 5 seconds and allow the coating to set at room temperature for 2 hours. Strip the coating from one side of the panel and bend at 180°F over a zero radius (fold back upon itself). Any evidence of cracking of the compound should be cause for rejection of the remaining material in the dipping tank.

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5.1.12.3 All parts coated with hot dip strippable protective compound are given a final visual inspection for continuous and acceptable coating before packing. Although the coating offers mechanical protection, if static loading is in excess of 30 pounds per square inch, cushioning must be used to reduce the load concentration. Cushioning of larger parts may be accomplished by overdipping the supporting areas with strippable protective compound or by overwrapping the dipped part with acceptable cushioning material. Packaged parts may be periodically checked by "rough handling" and "cyclic exposure" tests (see 7.2).

5.2 Procedure for use of cold-dip strippable compound.

5.2.1 Description of method. The method is effected by coating the prepared item with material conforming to MIL-P-45021. After solvent evaporation the item needs no further wrapping and is ready for packaging, since there is no exudation of oil (see figures 2, 4 and 5).

5.2.2 Static load (see 5.1.2)

5.2.3 Dipping equipment.

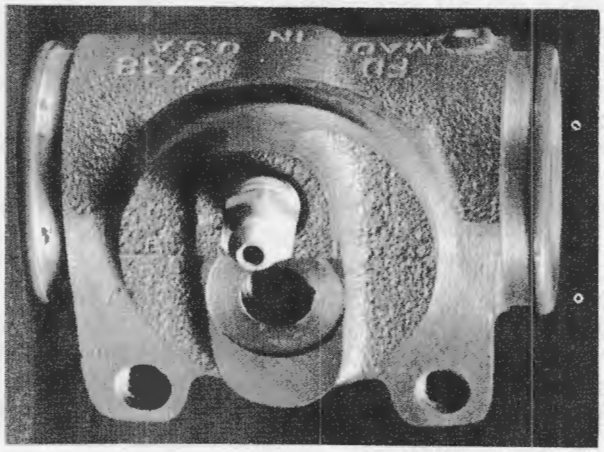
5.2.3.1 Because of the risk of igniting solvent vapors, open flames or any electrical equipment which might produce sparks should be avoided. Any motor driving a dipping machine, circulating pump, or any other nearby equipment should be of the explosion proof type. Adequate ventilation should be provided but no drafts should blow directly on the tanks.

5.2.3.2 All dipping and storage tanks should be evenly heated on all sides and the bottom. Tanks with closed jackets heated by circulating oil are the most satisfactory. When not in use, the tanks should be provided with a tight cover so that no evaporation loss occurs.

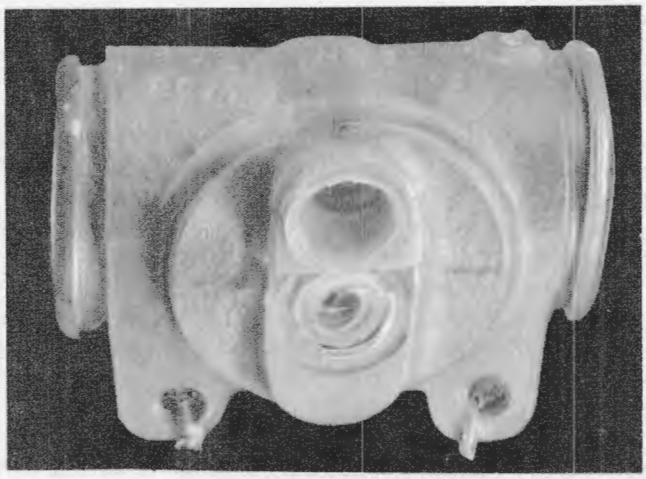
5.2.3.3 For small scale dipping, portable cans holding 1 to 5 gallons may be used. These can be heated electrically by means of a resistance wire or heater bands around the can, and controlling the temperature by means of a variable voltage control on the electrical supply.

Due care to provide thermal and electrical insulation around the outside of the container must be taken. When not in use, the tanks can be stored in a steam heated oven and kept at approximately the dipping temperature.

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4a. Brake cylinder after being dip coated per MIL-P-45021, humidity tested for thirty days and stripped.

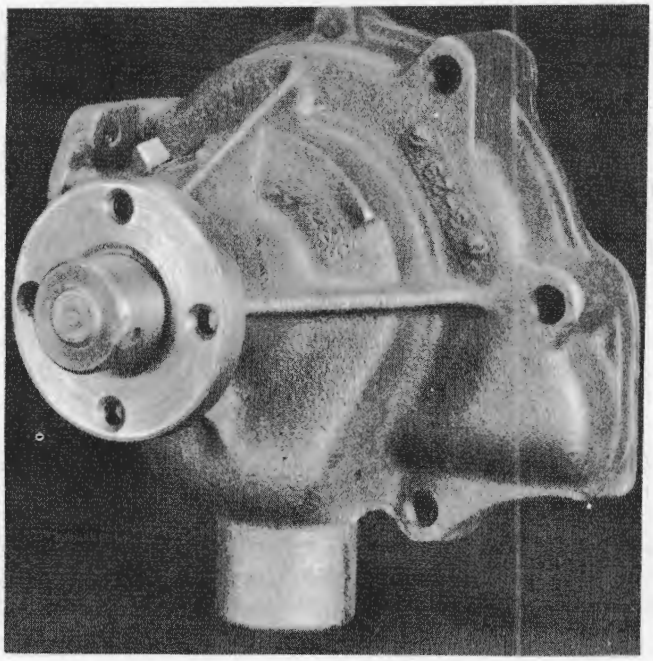


4b. Dipped film as stripped from brake cylinder.

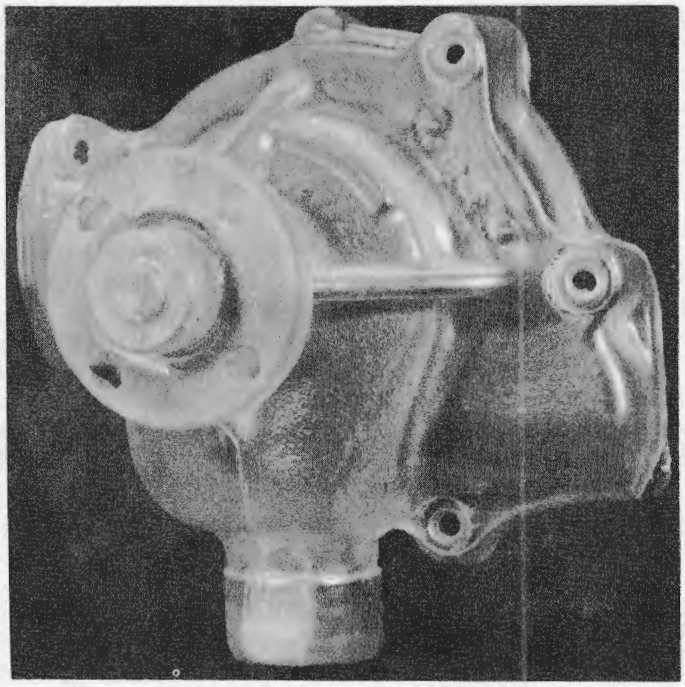
Figure 4. Brake wheel cylinder and stripped dip coating.

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5a. Pump after being spray coated per MIL-P-45021, humidity tested for thirty days and stripped. Note absence of corrosion.



5b. Spayed film as removed from pump.

Figure 5. Pump and stripped, spray coating.

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5.2.3.4 For large scale work, tanks similar to those used for hot-dip strippable compounds may be used. These tanks are described in MIL-T-13291.

5.2.4 Cleaning of items (see 5.1.4)

5.2.5 Suspension of items (see 5.1.5)

5.2.6 Material control.

5.2.6.1 In order to obtain a uniform product, it is necessary that both viscosity and "gel temperature" be maintained at constant values. To maintain these at constant values, the solids content and solvent ratio must be controlled.

5.2.6.2 "Gel test". The gel temperature is the temperature at which the compound sets to a firm gel without solvent evaporation. It is determined by removing a small sample of the compound into a test tube, stoppering it with a cork stopper having a thermometer extending into the compound, and allowing it to cool slowly. It is stirred occasionally with the thermometer. The temperature at which it no longer flows is the gel temperature. The transition is sharp.

5.2.6.3 Gel temperature maintenance. For best results, the gel temperature should be maintained normally from 87 to 89°F, or about 5°F above ambient, when it is higher than the normal gel temperature. As solvents evaporate from the compound, the gel temperature rises and will eventually become so high that satisfactory dipping can no longer be accomplished. To bring the compound back to its normal gel temperature, (87 to 89°F) the special thinner used is so designed that the addition of two-thirds of a fluid ounce per gallon of compound per degree F above the normal gel temperature will return the compound to its normal gel temperature and also maintain the viscosity fairly constant. (The special thinner is a mixture containing 55% isopropanol and 45% xylol by volume.)

5.2.6.4 Thickness. Sufficient thickness for a satisfactory coating of strippable protective compound, cold dip, is 0.015 to 0.025 inch. Coating thickness is determined by the dipping temperature, the immersion time, the mass or heat capacity of the part and conditions of the compound. While a part is being dipped it takes some of the heat from the melted plastic, and the longer the duration of the dip, the closer the temperature of the part approaches the temperature of the plastic and the thinner the protective compound obtained. Conversely a massive part when dipped for the same interval will retain a temperature considerably lower than the temperature of the plastic, resulting in greater cooling action on the plastic during the "drain off" period. This results in a thicker protective coating.

5.2.7 Plugging of cavities. The cold dip strippable compound will effectively bridge openings up to one inch in diameter. For items with openings greater than one inch, it is desirable to use a polyethylene plug before dipping. These plugs are described in MIL-C-5501. Irregular shaped openings may also be plugged with chunks of gelled compound.

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5.2.8 Dipping procedure.

5.2.8.1 Single dip. The actual dipping technique must be worked out for each article to be dipped. In general, the article should be dipped smoothly but fairly rapidly into the compound, care being taken not to entrap any air. The object should then be withdrawn relatively rapidly at first, slowing down toward the end and allowing the coating to drain 5 to 10 seconds before removing the object from the surface of the compound. After the compound has gelled the drainings can be cut from the object. Heavier coatings can be applied by repeated dippings (see figure 4).

5.2.8.2 Two step dip (see 5.1.8.2)

5.2.8.3 Cold-dip spray. Cold dip strippable compound can also be applied by the spray technique by mixing the following at 120°F:

1000 grams of cold dip strippable compound
1670 ml of xylol
80 ml isopropyl alcohol
750 ml isoamyl alcohol

The article is sprayed using a suitable spray gun with an E fluid tip and a 765 air cap with an air pressure of 30 p.s.i. The gun is held at an angle of 90 degrees and 8 to 10 inches from the object to be sprayed. The wet thickness of the films are such that films are produced which are 0.020 to 0.025 inch in thickness on drying (see figure 5). The number of passes required to produce this thickness will require some experimentation and will depend largely on the object to be coated.

5.2.9 "Setting-up" of the compound.

5.2.9.1 The compound can be dried completely at room temperature. Most of the solvent is removed after the first hour. For complete curing, the articles should be allowed to dry overnight at room temperature before packing for shipment. Initial drying and thorough gelling of the compound must be carried out at room temperature for the first 30 minutes, at which time the temperature can be raised. If accelerated drying is desired final curing may be carried out in a gently moving current of warm air at a temperature not greater than 130°F.

5.2.9.2 For multiple coatings at least ten minutes should be allowed between dips. It is best to make the interval as long as possible to insure the evaporation of the solvent from one dip, before the next dip is carried out.

5.2.10 Depreservation (see 5.1.10 and figure 5).

5.2.11 Inspection standards. All parts coated with cold dip strippable protective compound are given a final visual inspection for continuous and acceptable coating before packing. Packaged parts are periodically checked by the "rough handling" and "cyclic exposure" tests.

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6. USES AND APPLICATIONS

6.1 Hot-dipping strippable compound.

6.1.1 In general the use of hot-dip compound, MIL-P-149, is recommended because it is economical and easier to apply than the cold-dip type. No extensive amount of equipment is entailed, and the application is a quick single operation. It has the added advantage of the elimination of volatile solvents and the accompanying flammability and toxicity hazards. 28

6.1.2 Compound shall by type II of MIL-P-149, unless otherwise specified by the procuring agency. Type II is preferable when coating parts that will be exposed to exterior storage, and where transparency is necessary for complete visual inspection and identification of the coated parts.

6.1.2.1 Type I has the advantage of lower cost and offering a second source of supply in case of emergency. It may be used for indoor storage (see sections 4.1 to 4.4).

6.1.3 The high application temperature of hot-dip compounds limits its application to parts that are not to be preheated prior to coating, since this will result in thin coatings that will not strip satisfactorily. If a vapor degreaser has been used to clean parts, the parts must be cooled below 120°F before the dip can be applied.

6.1.4 The method is further limited to parts that have no deep recesses or intricate structure which would prevent the complete removal of the plastic. Articles which would be damaged by oil or by temperatures up to 320°F, for example electric coils, fuses, fabrics, paper etc. are not suitable for preservation by this method. Parts most adaptable to this method of preservation are those with plain continuous surfaces, such as connecting rods, engine valves, gears, shafts, yokes, cam shafts, bearing cups etc. Assemblies will not ordinarily be preserved by this method unless the design or contour is simple.

6.2 Cold-dipping strippable compound. Cold-dip strippable compound is intended specifically for application at moderate temperature (115-120°F) to parts containing heat sensitive materials. It is also intended for use in coating unmounted optical lenses and elements for physical protection and to render them resistant to fungal attack.

6.3 The strippable compounds provide a coating which permits easy handling and identification of the item without requiring unwrapping and exposure of the item to the weather or fingerprints. This packaging method is particularly applicable to items which are easily damaged by rust or corrosion such as gages, tools and machined parts.

6.4 Other packaging methods. Other packaging methods as described in MIL-P-116 may be cheaper and provide adequate protection for the item in question.

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7. METHODS OF TESTING AND EVALUATION

7.1 Testing of strippable compounds. The strippable compounds and raw materials may be tested by the applicable specifications as follows: MIL-P-149, MIL-P-45021, and ASTM Standards.

29 7.2 Testing of packages. The packages may be subjected to rough handling, cyclic exposure, or other tests as specified in accordance with MIL-P-116 and Fed. Test Method Std. No. 101.

8. MISCELLANEOUS

8.1 Suitable adhesives for labeling parts coated with plastic coating compound, strippable. Lacquers meeting MIL-L-10287, type II, lacquer, cellulose nitrate and thinner are superior to all adhesives that have been investigated for this purpose. Lacquers based on 1/2 second nitrocellulose are superior to those based on 30 to 35 centipoises. Nitrocellulose lacquers plasticized with nondrying liquid resin are superior to those with dibutyl phthalate, blown castor or abalyn. The resin reduces to a minimum the amount of bleeding of the oils and waxes in the label. Ditto paper is superior to bond paper as a labeling material.

8.2 Masking agents. Masking agents may be added to hot dipping compositions to minimize objectionable odors caused during heating above 350°F. For this purpose, Fritzche Brothers No. 23906, Dodge-Alcott Cedar wood oil and Syn Fleur CB #6 appear to be very effective. A concentration of masking agent between 0.025 and 0.25% based on the weight of the cellulose ester, is usually sufficient. The masking agent is added during the final stages of melt preparation, just prior to casting and cooling.

8.3 Colored coating. Colored, transparent, ethylcellulose coatings may be obtained by using oil soluble dyes. Only those which are heat and light stable should be used. They are added directly to the hot dip or may be predissolved in solvent and added to the melted hot dip compound.

8.4 Opaque coatings. Opaque coatings of any color may be obtained by the use of pigments. Because of the thickness of the coating, opacity may be obtained at lower pigment to vehicle ratio than is used in conventional lacquers. Approximately a 1-10 pigment to vehicle ratio, when titanium dioxide is used, will give opacity with a coating thickness of 15 mils. Other pigments should be used in greater or lesser amounts, depending on their relative hiding power.

8.5 Colored and pigmented cold dip coating. Tenite II supplied by Eastman Chemical Products, Inc. may be used to color cold dipping compounds.

Custodians:
Army - MR
Navy - SH
Air Force - 11

Preparing activity:
Army - MR

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C. Specifications and Standards

26. MIL-P-116 - Preservation, Methods of

27. MIL-A-148 - Aluminum Foil

28. MIL-P-149 - Plastic Coating Compound, Strippable (Hot Dipping) 31

29. MIL-T-3530 - Treatment, Mildew Resistant and Water Repellent, for Thread and Twine

30. MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal

31. MIL-C-6799 - Coating, Sprayable, Strippable, Protective

32. MIL-L-12087 - Lacquer, Cellulose Nitrate for Identification of Small Arms Ammunition (Type II)

33. MIL-C-11796 - Corrosion Preventive Compound, Petrolatum, Hot Application

34. MIL-S-12091 - String, Water Repellent, Non-Wicking (for Use with Strippable Protective Compounds)

35. MIL-C-16555 - Coating Compound, Strippable, Sprayable

36. MIL-T-13291 - Tanks, Hot-Dip, Directly and Indirectly Heated

37. MIL-C-15074 - Corrosion Preventive, Fingerprint Remover

38. MIL-P-45021 - Plastic Coating Compound, Strippable, Cold Dipping (120°F)

39. P-C-111 - Carbon Removing Compound

40. VV-S-190 - Sealing Compound (Dipcoat)

41 Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials; Test Procedures

42. ASTM Standards:

Method D817 - Testing Cellulose Acetate Butyrate
Method D914 - Testing Ethyl Cellulose

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